	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 1

LabTitle: Basic logic Gates and Verification of their Truth Tables

Student Name: Muhammad Waqas **Reg. No:**240756

Objective: To demonstrate the input and output relationship of 2 input AND, OR, NOT, NAND, NOR, XOR gates

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____

Lab Tasks:

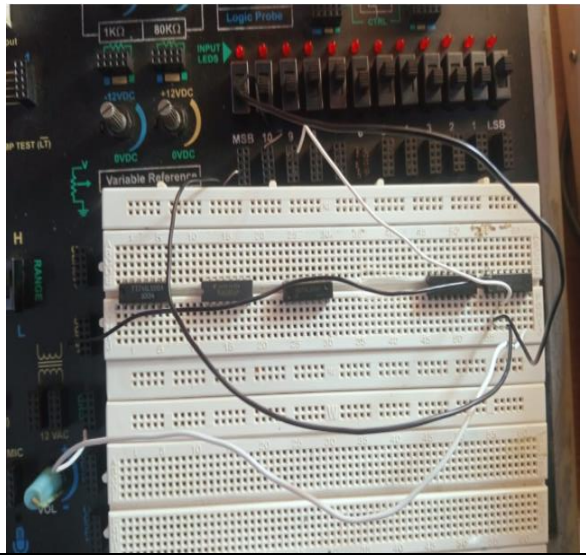
1. **Complete the truth table for AND Gate and verify using the required IC.**

IC Used:

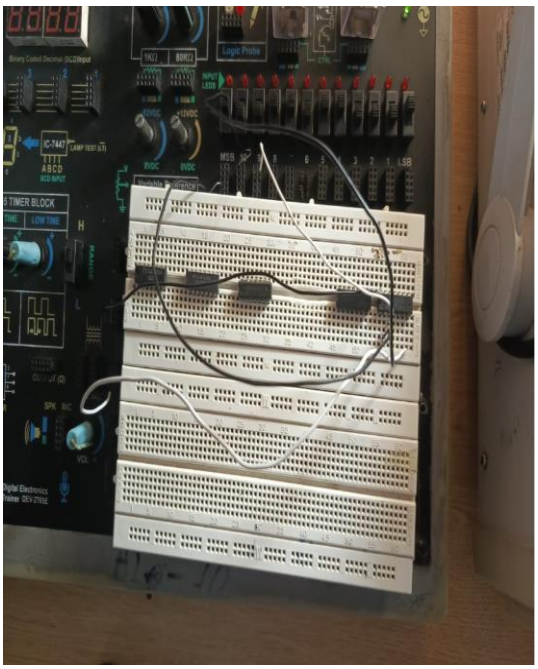
A=0 B=0 Y=0



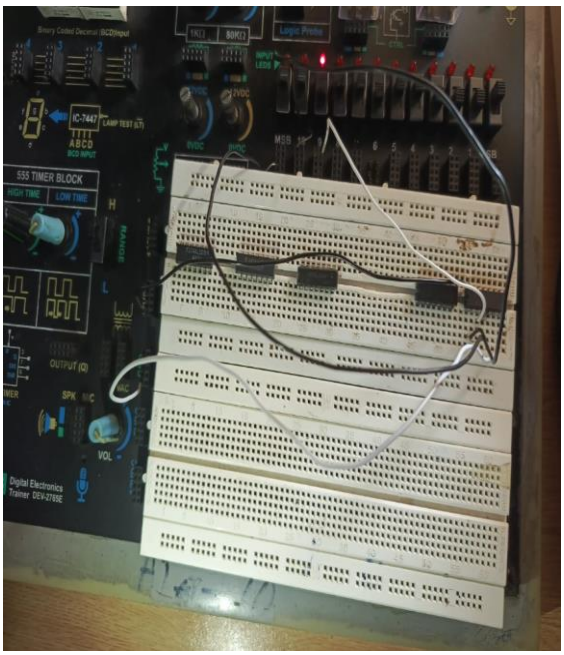
A=0 B=1 Y=0



A=1 B=0 Y=0



A=1 B=1 Y=1



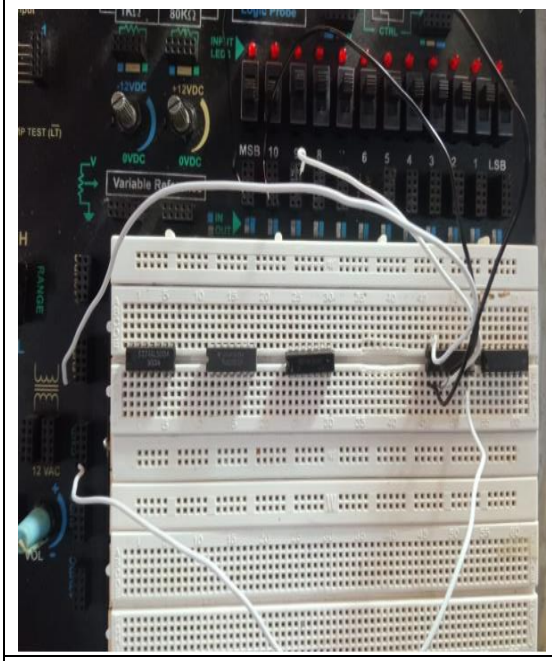
Truth Table:

A	B	Y
0	0	0
0	1	0
1	0	O
1	1	1

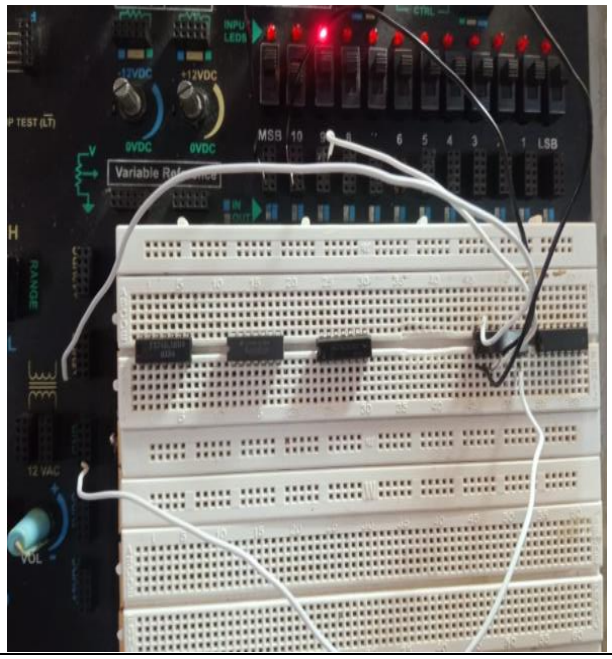
2. **Complete the truth table for OR Gate and verify using the required IC.**

IC Used:

A=0 B=0 Y=0



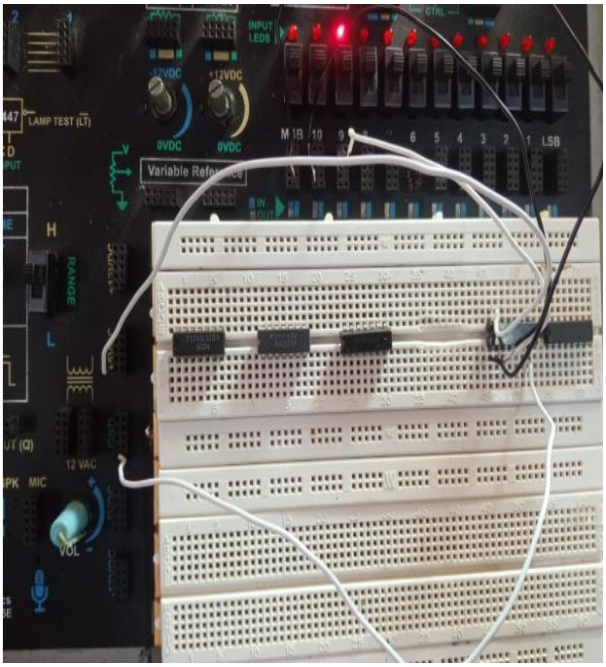
A=1 B=0 Y=1



A=0 B=1 Y=1



A=1 B=1 Y=1

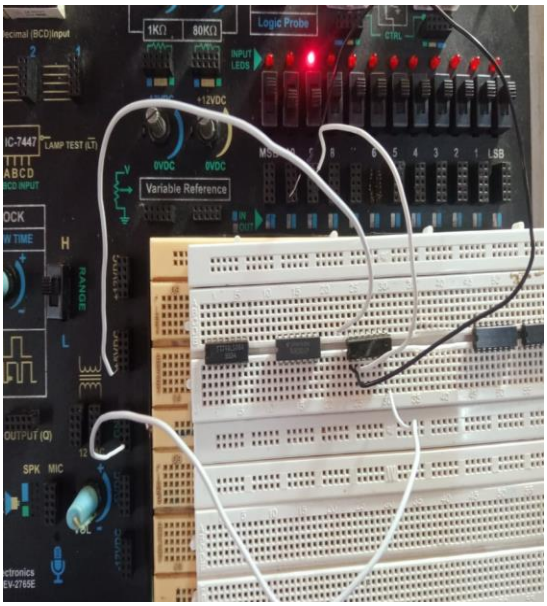
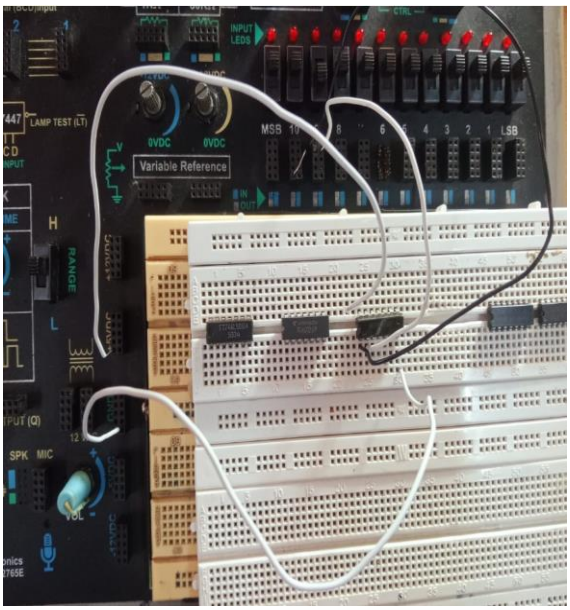


Truth Table:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

3. **Complete the truth table for NOT Gate and verify using the required IC.**

IC Used:

A=0 Y=1	A=1 Y=0
	

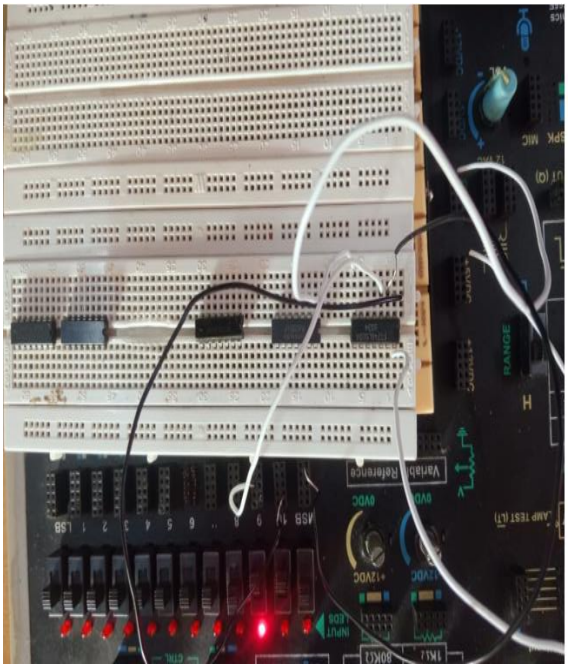
Truth Table:

A	Y
0	1
1	0

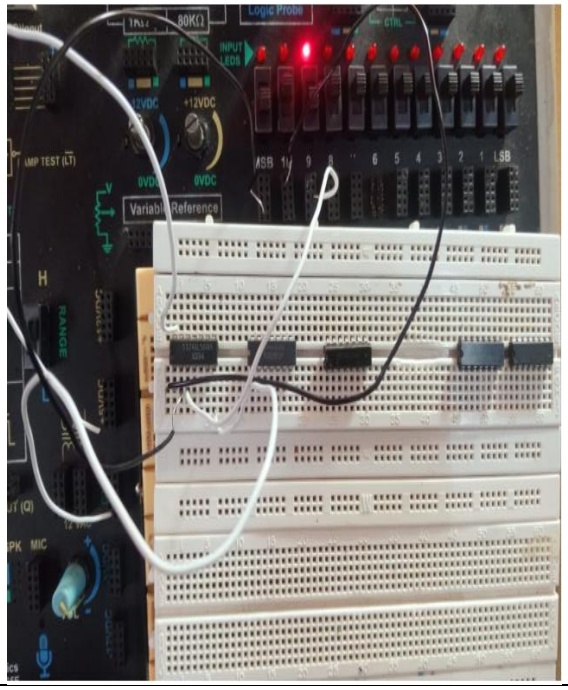
4. **Complete the truth table for NAND Gate and verify using the required IC.**

IC Used:

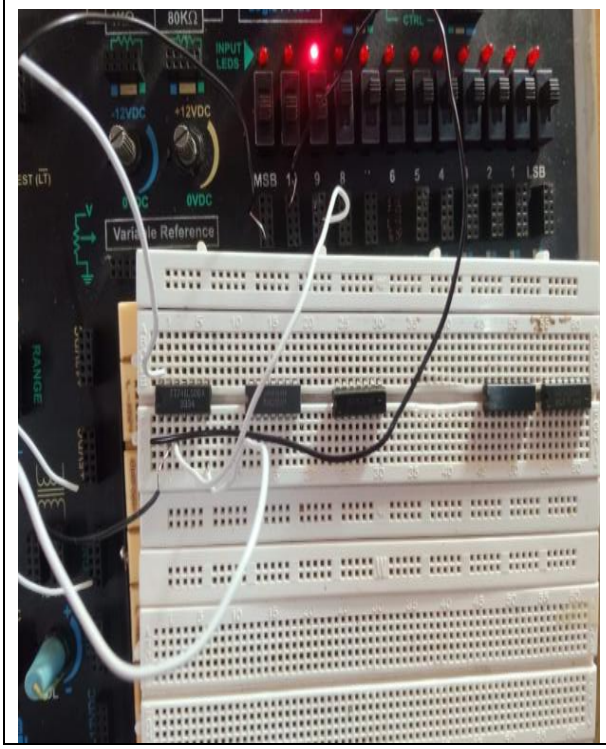
A=0 B=0 Y=1



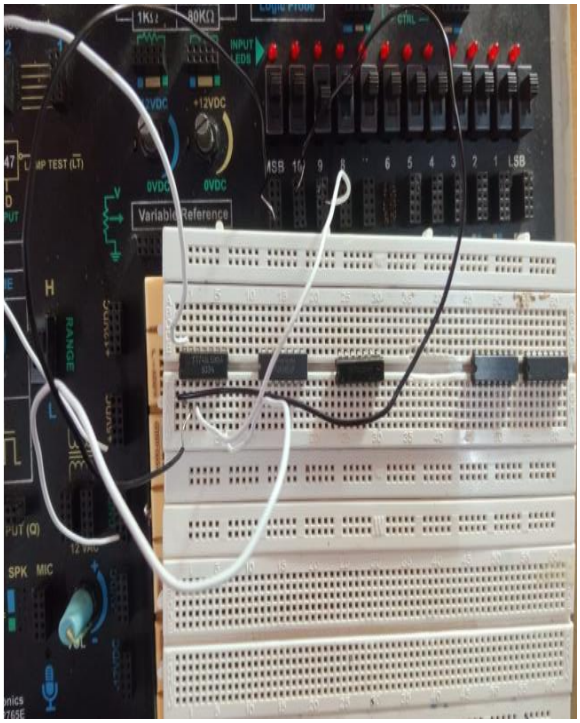
A=0 B=1 Y=1



A=1 B=0 Y=1



A=1 B=1 Y=0



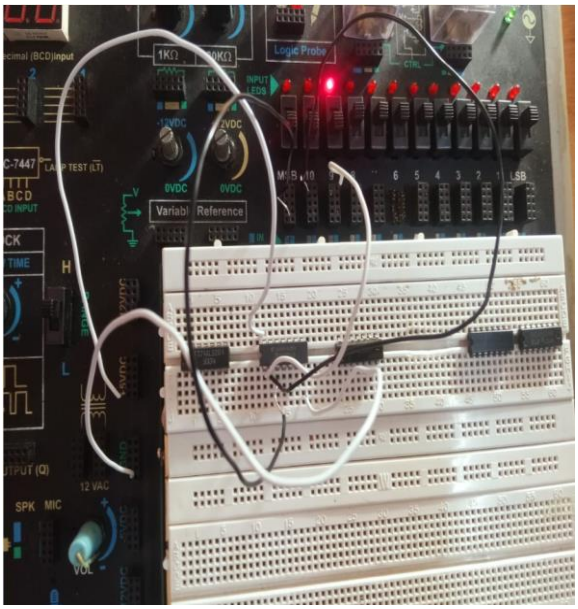
Truth Table:

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

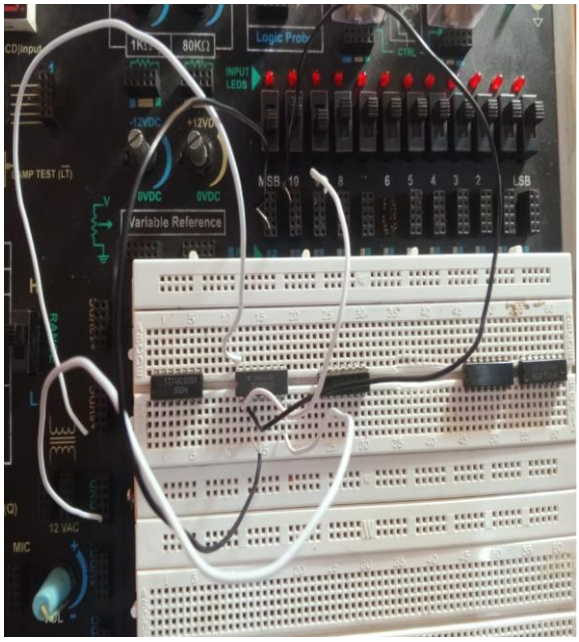
5. **Complete the truth table for NOR Gate and verify using the required IC.**

IC Used:

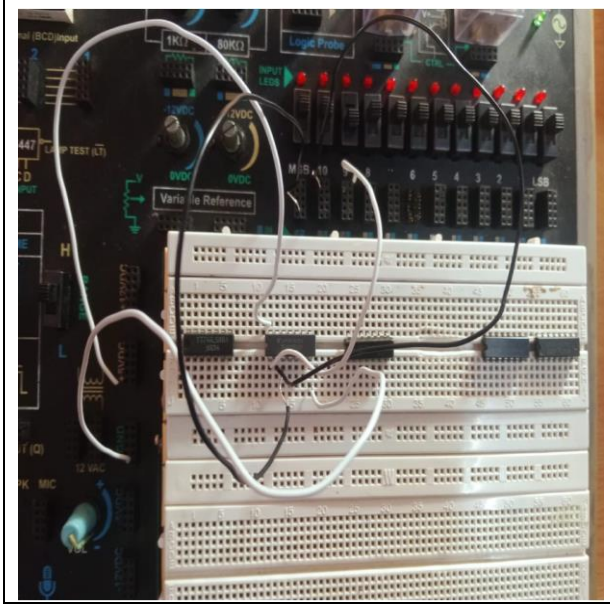
A=0 B=0 Y=1



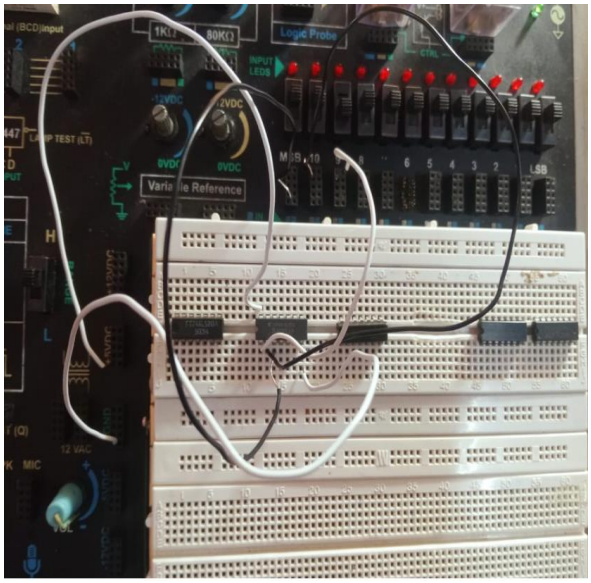
A=0 B=1 Y=0



A=1 B=0 Y=0



A=1 B=1 Y=0



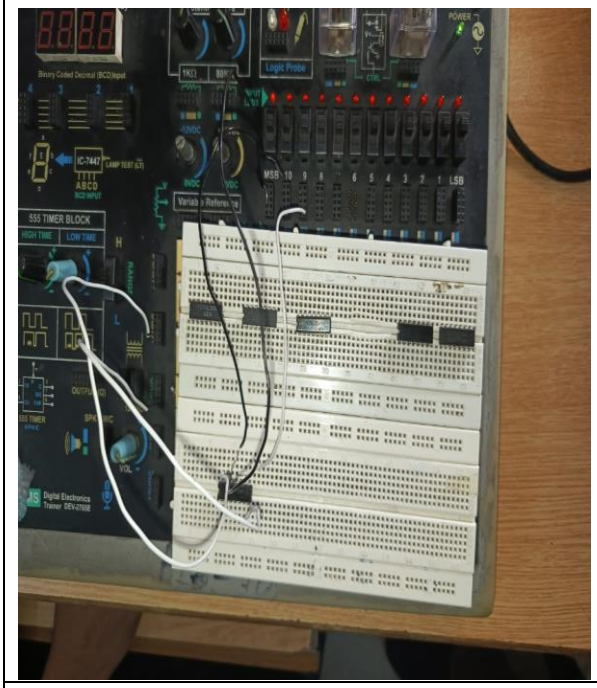
Truth Table:

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

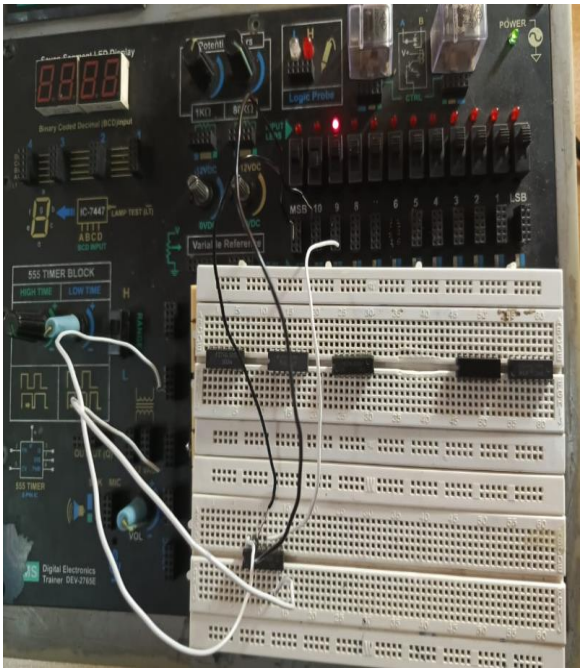
6. **Complete the truth table for XOR Gate and verify using the required IC.**

IC Used:

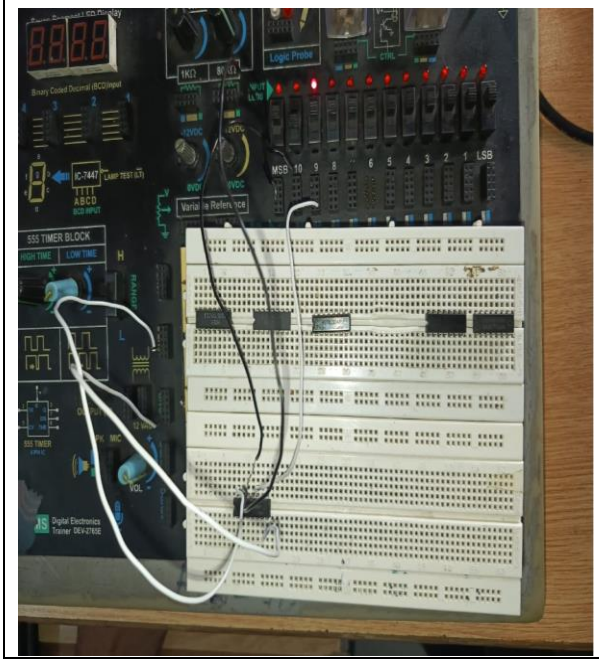
A=0 B=0 Y=0



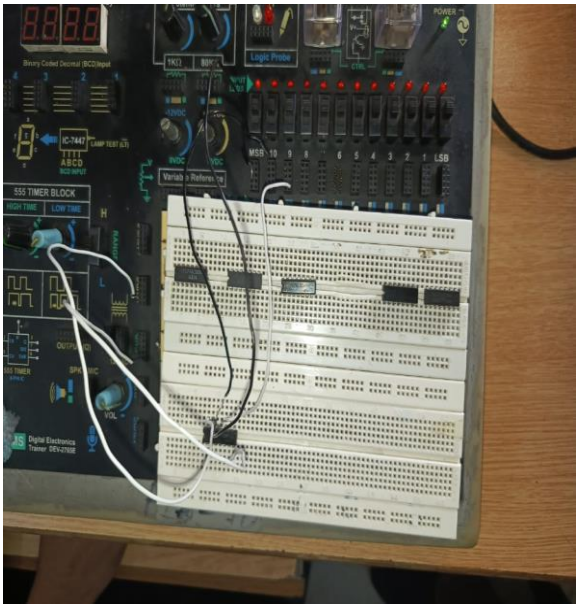
A=0 B=1 Y=1



A=1 B=0 Y=1



A=1 B=1 Y=0

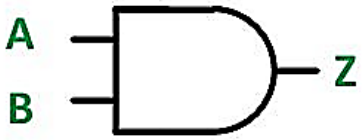


Truth Table:

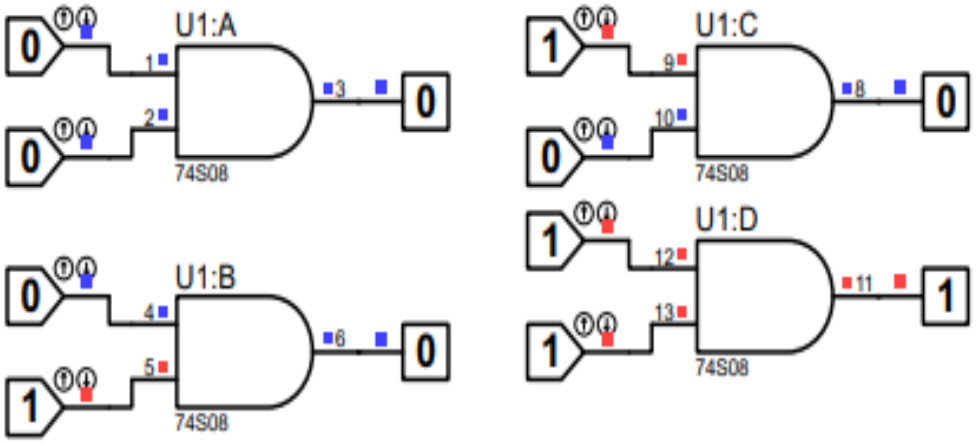
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Simulation Activities:

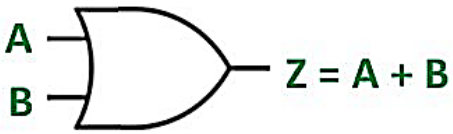
1. **Simulate the given circuit for AND gate in Proteus and verify its truth table.**



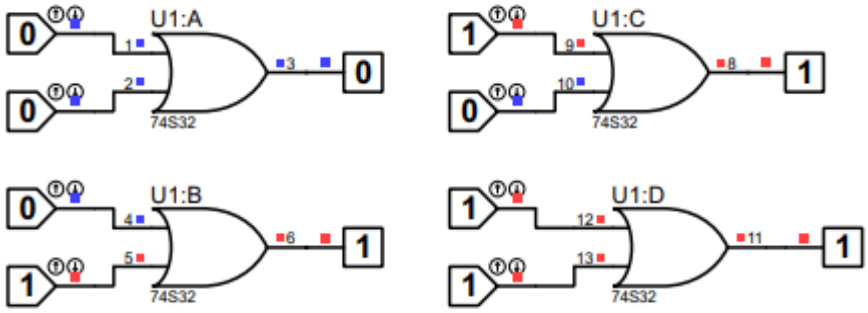
Simulation:



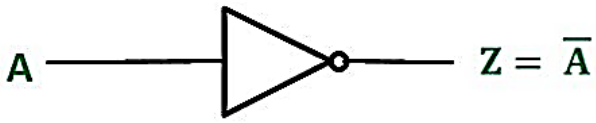
2. **Simulate the given circuit for OR gate in Proteus and verify its truth table.**



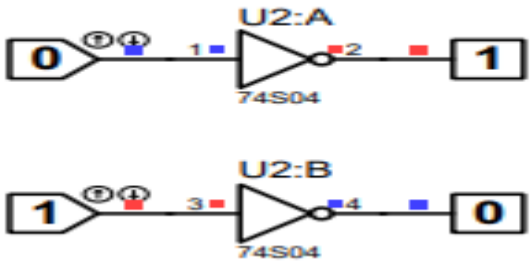
Simulation:



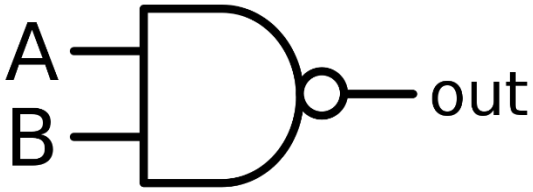
3. **Simulate the given circuit for NOT gate in Proteus and verify its truth table.**



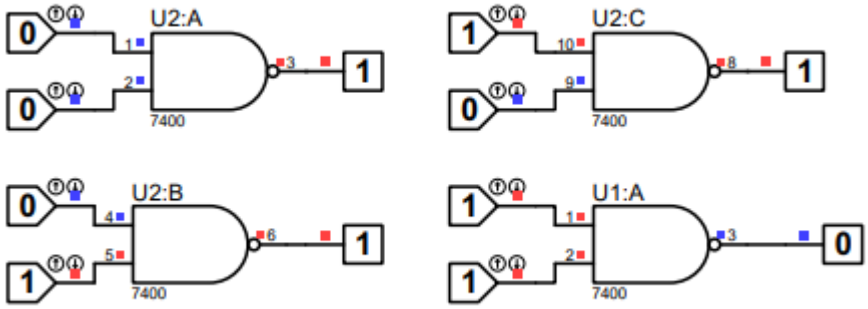
Simulation:



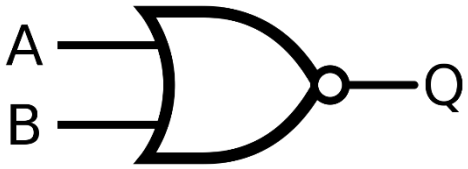
4. **Simulate the given circuit for NAND gate in Proteus and verify its truth table.**



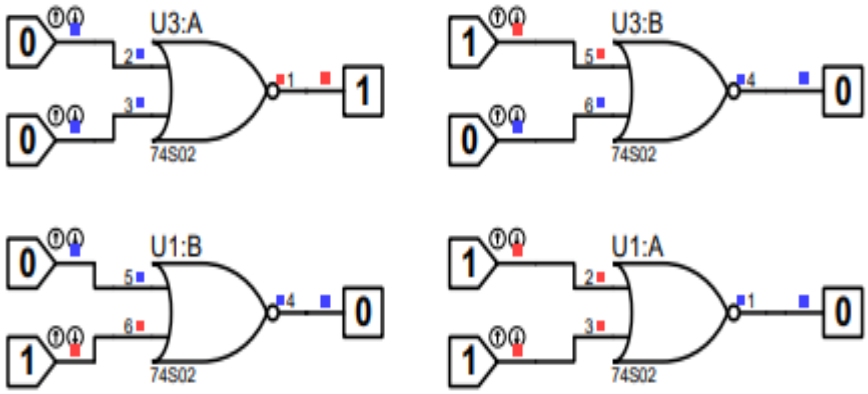
Simulation:



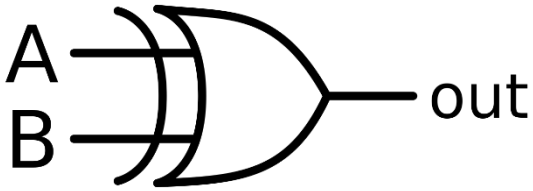
5. **Simulate the given circuit for NOR gate in Proteus and verify its truth table.**



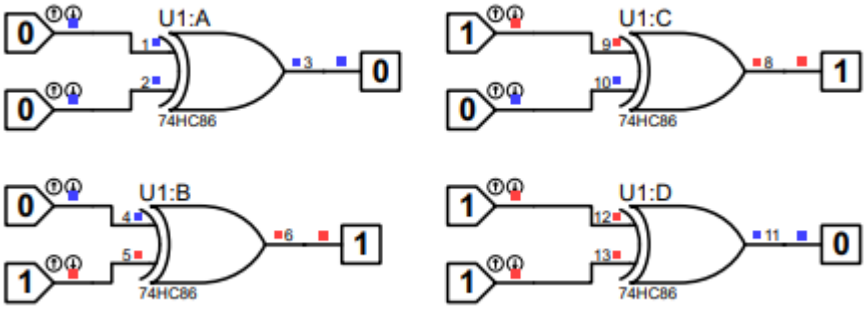
Simulation:



6. **Simulate the given circuit for XOR gate in Proteus and verify its truth table.**



Simulation:



Conclusion:

- Logic gates are the fundamental building blocks of all digital circuits, performing basic logical operations.
- The three basic gates are AND, OR, and NOT, which form the basis for more complex circuits.
- NAND and NOR gates are "universal gates" because any other logic gate can be built using only one type of these gates.
- Logic gates are manufactured on a single piece of semiconductor material as Integrated Circuits (ICs), allowing for the creation of compact and powerful devices.
- Specialized gates like XOR and XNOR are used for specific functions such as binary addition and data comparison.
- The operation of all logic gates is based on the rules of Boolean algebra, where inputs and outputs are binary values (0 or 1).